

CPM — Conditional Probability Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Probability & Uncertainty Standard (PPUS)

Operational Layer: Prediction Probability & Uncertainty Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Conditional Probability

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AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Conditional Probability

Canonical Definition

Conditional Probability Module (CPM) defines the governance conditions under which a probability remains explicitly associated with the assumptions, scenarios, states, events, dependencies, thresholds, or other conditions upon which its predictive meaning materially depends.

CPM governs how conditional probability is identified, represented, bounded, qualified, compared, updated, and preserved without allowing the condition to disappear while the probability value remains.



transforms:

Governed Probability Representation + Governed Predictive Uncertainty Profile + Governed Confidence Representation + Governed Predictive Range Representation + Applicable Predictive Conditions

into:

Governed Conditional Probability Representation

Its foundational principle is:

A probability that depends upon a condition changes meaning when the condition changes.

Operational Role

CPM is the fifth and final internal governance space of the Prediction Probability & Uncertainty Standard (PPUS) .

PRPM establishes probability meaning.

PUNM establishes uncertainty.

CFRM establishes confidence.

PRGM governs ranges and intervals.

CPM establishes whether the probability depends upon specific conditions and ensures those conditions remain part of the predictive representation.

The focused operational question is:

Under what conditions does this probability hold, and what happens to its meaning when those conditions change?

The primary output is:

Governed Conditional Probability Representation

which completes the PPUS internal governance process.

Internal Module Flow

Probability Representation → Predictive Uncertainty → Confidence Representation → Predictive Range Governance → Conditional Probability

Current Position: Conditional Probability

Conditional Probability

Conditional Probability expresses the probability of a predicted event or state under one or more defined conditions.

Canonical form:

$P(X | Y)$

where:

X = predicted event or state

and:

Y = applicable condition.

The condition is part of the predictive meaning.

Conditional Probability ≠ Unconditional Probability

Example:

80% probability of project completion by December if financing is approved by June

is not equivalent to:

80% probability of project completion by December.

Removing the financing condition changes the prediction's probability meaning.

Condition Identity

Every material condition should remain identifiable.

A condition may refer to:

assumption; scenario; external event; system state; operational state; regulatory state; market condition; behavior; threshold; dependency; prior event; resource availability; temporal state.

Condition Source

The origin of the condition should remain traceable.

Possible sources include:

PAS-governed Predictive Assumption; PBAS-governed information; ORA-governed Operational Reality signal; scenario definition; methodological requirement; system constraint; external dependency.

Condition ≠ Assumption by Default

Some conditions are Predictive Assumptions.

Others are:

observed states; scenario parameters; external events; methodological constraints.

Condition ≠ Predictive Assumption by Default

Condition Scope

A condition should identify what part of the prediction it affects.

It may affect:

event probability; magnitude; timing; range; confidence; feasibility; applicability.

Condition State

A condition may occupy states such as:

Holding

Conditionally Holding

Uncertain

Under Challenge

Breached

Not Yet Determined

No Longer Applicable

Condition State ≠ Prediction State

A condition may change state without automatically changing the overall prediction state.

PMRS governs prediction-level review and revision.

Single-Condition Probability

A probability may depend primarily on one condition.

Example:

$P(\text{Project Completion} \mid \text{Permit Approved})$

Multi-Condition Probability

A prediction may depend upon several simultaneous conditions.

Example:

$A \wedge P(\text{Project Completion} \mid \text{Permit Approved Financing Available Supplier Active})$

The conditions should not be silently reduced to one generic statement such as:
assuming normal conditions.

Condition Set

A Condition Set represents the governed collection of conditions applicable to one probability representation.

Condition Interaction

Conditions may interact.

Example:

Financing availability may have little effect if the supplier fails.

Therefore conditional probability may depend on combinations of conditions rather than isolated conditions.

Independent Conditions

Conditions should not be assumed independent without sufficient basis.

Multiple Conditions $\neq$ Independent Conditions

Dependent Conditions

One condition may influence another.

Example:

Regulatory approval may affect financing availability.

Such dependencies may materially change the conditional probability structure.

Nested Conditionality

A prediction may contain nested conditions.

Example:

Probability of launch by Q4 is 75% if approval occurs by June, provided financing remains available.

The nested structure should remain identifiable.

Conditional Chain

A Conditional Chain may be represented as:

Condition A $\rightarrow$ Condition B $\rightarrow$ Probability State C

where the probability depends upon sequential states.

Scenario-Conditional Probability

A probability may be defined within a scenario.

Example:

Base Scenario: 70%

Adverse Scenario: 35%

Favorable Scenario: 88%

Each value remains conditional upon its scenario definition.

Scenario Probability $\neq$ Unconditional Probability

Scenario probabilities should not be collapsed into one number unless a separate methodology governs the composition.

Scenario Weighting

Where scenarios are weighted, the weighting methodology should remain explicit.

Scenario Weight $\neq$ Scenario Probability by Default

State-Conditional Probability

Probability may depend upon a system or operational state.

Example:

Failure probability: 5% under normal load

Failure probability: 28% under sustained overload

Threshold-Conditional Probability

A prediction may apply only when a threshold is crossed.

Example:

Probability of default rises to 40% if debt ratio exceeds 6.0x.

The threshold is part of the probability meaning.

Temporal Conditionality

A condition may be time-bound.

Example:

Probability = 70% if approval occurs before 30 June.

The condition may cease to apply after that date.

Temporal Condition Expiry

Expired conditions must not remain attached as though still current.

Condition Validity Window

Where relevant, each condition should carry a period during which it is applicable.

Condition Persistence

Some conditions are assumed to persist.

Example:

if current regulation remains unchanged

The persistence requirement should remain explicit.

Condition Breach

A Condition Breach occurs when a condition required by the represented probability no longer holds.

Condition Breach $\neq$ Prediction Error

The predicted event may still occur after a condition is breached.

The original conditional probability may simply no longer be applicable.

Conditional Probability Validity

A conditional probability remains valid only while the conditions supporting it remain sufficiently applicable.

Valid Conditional Probability at State A $\neq$ Valid Conditional Probability at State B

Condition Change

A material condition change may require:

probability reassessment; uncertainty reassessment;

confidence reassessment; range reassessment; prediction review.

CPM identifies the effect.

PMRS owns prediction-level lifecycle revision.

Condition Change $\neq$ Automatic Prediction Revision

Conditional Probability Update

Where conditions change, a new probability representation may be established.

Example:

Initial:

$P(X | A) = 70\%$

Later:

A no longer holds

New state:

$P(X | B) = 42\%$

The two probabilities should remain historically distinct.

Probability Update $\neq$ Historical Replacement

Current Conditional Probability $\neq$ Historical Conditional Probability

Conditional Probability Provenance

Each conditional probability should remain associated with:

original probability source; methodology; conditions; condition state; horizon; applicable prediction version.

Condition Provenance

The origin of the condition itself should remain traceable where material.

Implicit Condition

Some predictive systems rely on conditions that are never explicitly represented.

Examples:

regulatory stability; uninterrupted supplier access; unchanged customer behavior; continued system availability.

An implicit material condition creates governance risk.

Implicit Material Condition ≠ Governed Condition

Hidden Conditionality

Hidden conditionality occurs where a probability appears unconditional but actually depends on one or more unstated conditions.

Unconditional Appearance ≠ Unconditional Prediction

Default Condition

Systems may use default conditions.

Example:

assuming normal market conditions

The meaning of the default must remain defined.

Undefined Normality

A phrase such as:

under normal conditions

is insufficient where materially different interpretations are possible.

Normal Conditions ≠ Governed Conditions by Default

Condition Qualification

Conditions may require qualification where their state is uncertain.

Example:

Probability: 68%, conditional upon supplier continuity, currently assessed as uncertain.

Condition Confidence

A condition may itself have a confidence or support state.

This must remain distinguishable from event probability.

Probability of Condition vs Probability Conditional on Condition

These are different concepts.

Example:

$P(\text{Project Completion} \mid \text{Financing Available})$

is distinct from:

$P(\text{Financing Available})$

Probability of Condition $\neq$ Probability Conditional on Condition

Joint Probability

Where both condition and outcome probabilities are modeled, joint probability may be relevant.

Example:

$P(\text{Project Completion} \cap \text{Financing Available})$

This must not be confused with conditional probability.

Conditional Probability vs Joint Probability

$P(X \mid Y) \neq P(X \cap Y)$

Base Rate

Conditional probability may depend upon underlying base rates.

Where material, those base rates should remain identifiable through methodology.

Base-Rate Shift

Changes in the underlying population or environment may alter conditional probability.

Conditional Probability and Distribution Shift

A conditional probability established under one operating environment may become misleading under another.

Conditional Probability and Predictor–Reality Proximity

Where a predictive source has unusually close access to operational decisions or outcome-determining information, the conditions supporting its probability may differ materially from those of public-information predictors.

CPM does not determine Predictor–Reality Proximity itself.

It preserves relevant conditional dependencies where they affect probability meaning.

Outcome-Informed Conditions

Where a predictor has access to information materially closer to outcome determination, that information must not be silently treated as ordinary predictive conditions if it changes the nature of the probability.

Outcome Knowledge $\neq$ Predictive Capability

Condition and Reflexivity

A communicated prediction may itself influence one of its own conditions.

Example:

A forecast of product shortage may trigger stockpiling, changing actual demand.

Where material, continuing probability may become conditional upon the behavioral response induced by the prediction.

Prediction Observation $\neq$ Prediction Influence

Self-Altering Conditions

Where the prediction materially changes the environment it predicts, the original condition set may no longer remain valid.

Conditional Probability and Assumptions

PAS governs Predictive Assumptions.

CPM governs how probability depends upon those assumptions.

Conditional Probability and Prediction Basis

PBAS governs Prediction Basis.

CPM may use basis-derived conditions but does not re-govern the basis.

Conditional Probability and Uncertainty

PUNM governs uncertainty.

CPM preserves uncertainty concerning whether conditions will hold and how condition changes affect probability.

Conditional Probability and Confidence

CFRM governs confidence.

Confidence may vary across condition states.

Example:

$P(X | A) = 75\%$, High Confidence

$P(X | B) = 75\%$, Low Confidence

Same probability does not imply same confidence.

Conditional Probability and Ranges

PRGM governs predictive ranges.

A range may itself be conditional.

Example:

Expected revenue: CHF 100–120 million if capacity expansion is completed.

Conditional Probability and Feasibility

PFES later governs Prediction Feasibility.

Conditional probability does not determine whether the conditioned outcome is actually feasible.

Conditional Probability and Collective Prediction

CPGS governs Collective Prediction.

Where multiple predictors operate under different conditions, their probability estimates must not be combined as though they shared one condition set.

Same Event + Different Conditions $\neq$ Directly Comparable Probability by Default

Conditional Probability and Communication

PCRS later governs how conditions must be preserved when a prediction is communicated or relied upon.

CPM ensures that those conditions already exist as governed predictive information.

Conditional Probability and Revision

PMRS governs prediction monitoring and revision.

CPM establishes condition changes that may trigger that process.

Condition Trigger

A Condition Trigger is a defined condition-state change capable of requiring reassessment.

Examples:

threshold crossed; assumption breached; supplier fails; regulation changes; scenario shifts; financing expires.

Condition Trigger ≠ Revision Decision

Conditional Probability State

A conditional probability may occupy states such as:

Established

Conditionally Established

Condition Under Challenge

Condition Breached

Expired

Superseded

Unable to Determine

Expired Conditional Probability

A probability whose condition is no longer applicable should not remain represented as currently valid.

Superseded Conditional Probability

A new conditional probability may supersede an earlier one.

The earlier value must remain historically preserved.

Condition Conflict

Different systems may apply materially different conditions to apparently identical predictions.

Example:

System A:

70% if rates remain below 4%

System B:

70% assuming no recession

These values are not equivalent merely because the number and predicted event are the same.

Conditional Probability Conflict ≠ Numerical Conflict Only

The probability values may match while the conditions conflict.

Condition Completeness

A probability may be conditionally incomplete if one or more material dependencies remain unstated.

Conditionality Governance Gap

A Conditionality Governance Gap exists where:

a material condition is unidentified; condition scope is unclear; condition state is unknown; probability is displayed without required condition; condition dependency is hidden; scenario meaning is unclear; historical condition state is unavailable.

Conditional Probability Governance Requirements

CPM requires that:

1. every material conditional probability remain associated with a specific Governed Prediction; 2. the predicted event or state remain identifiable; 3. each material condition remain identifiable; 4. condition sources remain traceable where material; 5. conditions remain distinguishable from assumptions where applicable; 6. condition scope remain identifiable; 7. condition state remain representable; 8. condition state remain distinguishable from Prediction State; 9. multi-condition probabilities preserve the complete material condition set; 10. condition dependencies remain identifiable where material; 11. nested conditionality remain representable; 12. scenario-dependent probabilities preserve scenario identity; 13. scenario weighting remain distinguishable from scenario probability; 14. state-dependent probabilities preserve applicable system state; 15. threshold-dependent probabilities preserve applicable threshold; 16. temporal conditions preserve their validity window; 17. expired conditions not remain represented as current; 18. condition breaches remain identifiable; 19. condition breach remain distinguishable from Prediction Error; 20. condition changes remain capable of triggering reassessment; 21. condition changes not automatically be treated as prediction revision; 22. new conditional probabilities not overwrite historical representations; 23. conditional probability provenance remain traceable; 24. implicit material conditions be surfaced where reasonably possible; 25. hidden conditionality not be represented as unconditional probability; 26. undefined phrases such as "normal conditions" not substitute for materially required conditions; 27. probability of a condition remain distinguishable from probability conditional on that condition; 28. conditional probability remain distinguishable from joint probability; 29. base-rate or environment changes remain governable where material; 30. self-altering predictive conditions remain representable where material; 31. condition triggers remain distinguishable from revision decisions; 32. historical conditional probability states remain reconstructable.

Governance Logic

CPM applies the following governance logic:

Governed Probability Representation

Governed Predictive Uncertainty Profile

Governed Confidence Representation

Governed Predictive Range Representation

↓

Conditionality Requirement Determined

↓

Condition(s) Identified

↓

Condition Source / Scope Established

↓

Condition State Established

↓

Single / Multiple / Nested Condition Structure Determined

↓

Condition Dependencies Assessed

↓

Scenario / State / Threshold / Temporal Conditionality Assessed

↓

Condition Validity Window Established

↓

Conditional Probability Meaning Established

↓

Probability of Condition / Conditional Probability / Joint Probability Distinguished

↓

Condition Change / Breach / Expiry Triggers Established

↓

Conditionality Gaps Assessed

↓

Historical Condition and Probability State Preserved

↓

Governed Conditional Probability Representation

↓

PPUS Governance Complete

↓

PFES — Prediction Feasibility Standard

Where sufficient conditional governance cannot be established:

Conditionality Governance Gap → Clarification / Condition Disclosure / Qualification / Restricted Representation / Probability Reassessment / Unable to Determine

Minimum Implementation Framework

A minimum implementation of CPM must establish:

A. Prediction Reference

The applicable Governed Prediction.

B. Probability Representation Reference

The applicable Governed Probability Representation.

C. Condition Identifier

A distinguishable identifier for each material condition.

D. Condition Type

At minimum:

Assumption / Scenario / Operational State / External Event / Threshold / Temporal / Dependency / Regulatory / Market / Behavioral / Other

E. Condition Source

The origin of the condition.

F. Condition Definition

A clear representation of what must hold.

G. Condition Scope

The predictive element affected.

H. Condition State

At minimum:

Holding / Conditionally Holding / Uncertain / Under Challenge / Breached / Not Yet Determined / No Longer Applicable

I. Condition Dependency

Where the condition depends upon or influences another condition.

J. Condition Validity Window

Where temporal applicability matters.

K. Conditional Probability Representation

The probability applicable under the stated condition or condition set.

L. Scenario / State / Threshold Reference

Where applicable.

M. Change / Breach / Expiry Trigger

The condition-state change requiring reassessment.

N. Conditionality State

At minimum:

Established / Condition Under Challenge / Condition Breached / Expired / Superseded / Unable to Determine

O. Historical Condition / Probability State

The applicable historical record.

Governance Outputs

CPM may produce:

Governed Conditional Probability Representation; Conditional Probability Record; Condition Identifier; Condition Definition; Condition Type; Condition Source; Condition Scope; Condition State; Condition Dependency Map; Condition Set; Nested Conditionality Record; Scenario Conditional Probability; State Conditional Probability; Threshold Conditional Probability; Temporal Conditional Probability; Condition Validity Window; Condition Breach Record; Condition Expiry Record; Conditional Probability Update

Record; Condition Trigger; Conditionality Governance Gap; Conditional Probability State; historical condition state; historical conditional probability; Conditional Probability Version.

The primary governed output is:

Governed Conditional Probability Representation

which completes the internal governance process of the:

Prediction Probability & Uncertainty Standard (PPUS)

and becomes an input to:

PFES — Prediction Feasibility Standard.

Critical Distinctions

A Probability That Depends Upon a Condition Changes Meaning When the Condition Changes

Conditional Probability $\neq$ Unconditional Probability

Condition $\neq$ Predictive Assumption by Default

Condition State $\neq$ Prediction State

Multiple Conditions $\neq$ Independent Conditions

Scenario Probability $\neq$ Unconditional Probability

Scenario Weight $\neq$ Scenario Probability by Default

Condition Breach $\neq$ Prediction Error

Valid Conditional Probability at State A $\neq$ Valid Conditional Probability at State B

Condition Change $\neq$ Automatic Prediction Revision

Probability Update $\neq$ Historical Replacement

Current Conditional Probability $\neq$ Historical Conditional Probability

Implicit Material Condition $\neq$ Governed Condition

Unconditional Appearance $\neq$ Unconditional Prediction

Normal Conditions $\neq$ Governed Conditions by Default

Probability of Condition $\neq$ Probability Conditional on Condition

$P(X | Y) \neq P(X \cap Y)$

Outcome Knowledge $\neq$ Predictive Capability

Prediction Observation $\neq$ Prediction Influence

Same Event + Different Conditions $\neq$ Directly Comparable Probability by Default

Condition Trigger $\neq$ Revision Decision

Conditional Probability Conflict $\neq$ Numerical Conflict Only

Enterprise Application

CPM enables organizations to prevent one of the most dangerous forms of predictive simplification:

removing the conditions while keeping the probability.

This is especially important where predictions depend upon:

financing; regulation; supplier continuity; human behavior; market states; operational capacity; system state; scenario assumptions; external events;

timing thresholds.

A conditional probability can look operationally simple while depending upon a complex condition structure.

CPM keeps that structure visible.

Use Case 1 — The Condition Disappears From the Dashboard

A project forecasting model states:

80% probability of completion by December if financing is secured by 30 June.

A management dashboard later displays only:

Completion probability: 80%.

The financing condition disappears.

Executives continue relying on the 80% value even after the financing decision is delayed beyond June.

The critical governance question is:

Can the 80% probability remain valid when the condition that gave the number its meaning is no longer satisfied?

CPM requires the financing condition, validity window, condition state, and applicable conditional probability to remain linked.

Once the condition is breached or expires, the original 80% cannot continue to appear as an unconditional current probability.

Conditional Probability $\neq$ Unconditional Probability

Use Case 2 — Two Identical Probabilities Depend on Different Worlds

Two systems estimate:

70% probability that a factory expansion will be operational by Q4 2027.

System A means:

70% if regulatory approval is obtained by February.

System B means:

70% assuming current energy prices remain below a defined threshold.

A comparison dashboard presents both values as equivalent forecasts.

The critical governance question is:

Are two identical probabilities directly comparable when each is valid under a different condition set?

CPM requires each probability to preserve its condition identity, state, scope, validity window, and provenance.

The values may be numerically identical.

Their predictive meaning is not.

Same Probability Value + Different Conditions $\neq$ Same Conditional Probability

Canonical Boundary

CPM governs the identification, source, definition, scope, state, dependency, validity window, scenario relationship, threshold relationship, temporal relationship, nested structure, breach, expiry, update, trigger, provenance, state, versioning, and historical preservation of conditions upon which probability materially

depends. It does not govern Predictive Assumption identification itself, Probability Representation itself, Predictive Uncertainty itself, Confidence Representation itself, Predictive Range Governance itself, Prediction Feasibility, Collective Prediction composition, communication and reliance, prediction-level revision decisions, outcome determination, or predictive performance assessment.

CPM governs:

the conditions under which a probability retains its predictive meaning.

It does not determine:

whether those conditions will actually hold or whether the predicted outcome will occur.

Canonical Closing Statement

A probability without its condition can become a different prediction while keeping the same number.

CPM preserves the assumptions, states, scenarios, thresholds, and dependencies that give conditional probability its legitimate meaning.

If the condition changes, the probability must not pretend that nothing changed.

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